Smart Vault Security in Banks: A Quantum AI-Enabled Intrusion Detection System

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Abstract*—The rapid evolution of advanced persistent threats has exposed the limitations of traditional intrusion detection systems (IDS) in securing modern banking infrastructures. In this work, we propose a Quantum AI-Enabled Intrusion Detection System (QAID) designed for smart vault security, integrating quantum computing techniques with advanced artificial intelligence (AI). The system utilizes quantum feature encoding and quantum kernel methods to detect sophisticated, non-linear intrusion patterns in high-dimensional security data that classical machine learning- based IDS often overlook. Using quantum support vector machines (QSVM), QAID enhances anomaly detection, precision, and operational resilience against adversarial attacks. The architecture of the system, the mathematical framework, and the algorithm implementation are described in detail, followed by an evaluation of a real-world smart vault security dataset. The experimental results confirmed that QAID consistently outperformed conventional IDS techniques in terms of detection accuracy, robustness to high-dimensional data, and real-time operational capability. These findings highlight the potential of quantum AI-based approaches in strengthening cybersecurity frameworks for critical financial infrastructures.

Keywords—Quantum Computing, Smart Vaults, Banking Security, Intrusion Detection, Quantum Kernel Methods, Quantum Support Vector Machine.

I. Introduction

The introduction of smart vaults has changed traditional and modern cybersecurity techniques within the banking sector. These high-tech vaults integrate IoT devices, biometric identification, and advanced encryption algorithms to offer unparalleled security while posing a suitable challenge to standard security measures. Although these techniques offer enhanced protective measures, they remain beneficial for advanced breaches and cyberattacks as well. The classical machine learning-based systems lack the ability to capture intricate intrusion patterns due to their nonlinear dimensional data complexity. Traditional cyber intrusion detection systems based on machine learning fail to capture underlying and complex intrusion patterns. Recently, there have been new breakthroughs in quantum computing that have quantum machine learning (QML) as a subset, which deals with recognizing and differentiating patterns of anomalies. Quantum algorithms, like QSVM and Quantum SVMs, are capable of using quantum parallelism and entanglement to outperform classical algorithms for large dataset processing and analysis. This paper offers a revolutionary take by suggesting hybrid classical and quantum computer integration for developing an intrusion detection system where a smart vault security feature is encoded using a quantum feature map and classified with a quantum enhanced intruder detection system.

II. Implementation

In this section, we describe the implementation of our QAID , which is designed for smart vault security in banks. We describe the configuration of the environment, the fusion of quantum and classical components, data encoding, and the subsequent steps of kernel calculation and classification in a detailed manner.

A. Environment Setup

Quantum Frameworks: Quantum components are implemented using Qiskit and PennyLane. These frameworks allow us to simulate quantum circuits that perform feature encoding and kernel computations.

Classical Frameworks: We use Python along with libraries such as NumPy, SciPy, and scikit-learn. The QSVM classifier is implemented using the scikit-learn SVM with a custom kernel framework.

Hardware: Simulations are executed on a workstation with an Intel i7 processor, 32GB of RAM, and GPU acceleration where applicable.

B. Quantum Feature Encoding

Each sample of data X is mapped into a quantum state via a parameterized unitary operator $U(X)$. It achieves this mapping via a quantum circuit implementing rotation gates that depend on the value of the input feature. For instance:

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1) Algorithm 1: Quantum Feature Encoding using Qiskit

- Input: Classical feature value x
- Output: Quantum circuit encoding the feature as a rotation
- Begin
- Initialize a single-qubit quantum circuit qc.
- Apply a rotation gate $R_{y(x)}$ to qubit 0.
- Return the quantum circuit qc.
- End

C. Quantum Kernel Computation

It is numerically simulate the quantum states and calculate the inner product to get the kernel value:

2) Algorithm 2: Compute Quantum Kernel Value

- Input: Quantum circuits qc_1, qc_2
- Output: Quantum kernel value $K(qc_1, qc_2)$
- Begin
- Set backend as the statevector simulator.
- Execute qc_1 on the backend to obtain state1.
- Execute qc_2 on the backend to obtain state2.
- Compute the inner product between state1 and state2.
- Calculate Kernel value as the square of the absolute value of the inner product.
- Return the Kernel value.
- End.

D. Quantum SVM Classifier:

The QSVM classifier is achieved by constructing a specific kernel matrix based on the quantum kernel function. We then employ this matrix to train a Support Vector Machine (SVM):

3) Algorithm 3: Quantum Kernel-Based SVM Classification

- Input: Feature set X, labels y
- Output: Trained quantum kernel-based SVM model
- Define quantum_kernel (X, Y)
- Initialize a kernel matrix of size $|X| * |Y|$
- For each $x_i \in X$
- For each $y_j \in Y$
- Return the kernel matrix.
- Initialize training data X_{train}, y_{train}
- Create an SVM classifier with kernel = quantum_kernel
- Fit the SVM classifier to X_{train} and y_{train}
- End.

E. Real-Time Integration

In order to facilitate real-time intrusion detection, the QAID system is interfaced with the bank's smart vault security network via an API that collects data from:

- IoT sensors
- Biometric systems
- Access logs

The collected data are pre-processed (cleaned and normalized) and then fed into the QSVM encoding and classification pipeline. If the classifier detects an anomaly (an intrusion activity), an alert is generated and an appropriate response is triggered.

III. System and Methodology

A. System Architecture

The QAID system is made up of the following modules :

- Data Acquisition Module: Collects multimodal security data from sensors, biometric devices, access logs, and network traffic.
- Preprocessing Module: Cleans and normalizes the data and extracts relevant features.
- Quantum Feature Encoding Module: Maps classical data into quantum states using a parameterized unitary transformation.
- Quantum Kernel Computation Module: Computes a quantum kernel matrix that captures the similarity between data points.
- Classifier Module: Uses a quantum-enhanced SVM (QSVM) to classify normal versus anomalous access events.
- Alert and Response Module: Generates alerts and triggers response protocols when an intrusion is detected.

B. Quantum Feature Encoding

Each data sample $X \in R^d$ is encoded as a quantum system using an integrated operator $U(X)$. For instance, an amplitude encoding scheme is defined as:

$$\langle|\varphi_{up}(x)\rangle = \langle U(x)|0\rangle \tag{1}$$

where $\langle|0\rangle$ is the initial quantum state.

C. Kernel Method

Kernel in the middle of two data samples is defined as the squared inner product of their quantum state representations:

$$K(x_{i,}x_j) = |\langle\varphi(x_i)|\varphi(x_j)\rangle|^2 \tag{2}$$

This kernel function captures complex, high-dimensional *correlations* that *may be* missed by classical kernels. The resulting to train the classifier.

IV. Algorithm

It explains the algorithmic framework of the proposed Quantum AI Intrusion Detection (QAI-ID) system in this particular segment. The algorithm incorporates quantum feature encoding and quantum kernel methods to construct a QSVM-based classifier that differentiates between normal behavior and intrusion attempts in smart vault systems.

A. Quantum AI Intrusion Detection(QAI-ID) Algorithm

Quantum AI Intrusion Detection (QAI-ID)

Dataset $D = \{(x_i, y_i)\}_{i=1}^{N}$, New sample = x_{new}

Preprocess: normalize all feature vectors in D. each $x_i \in D$
Compute quantum state:

$$\langle|\varphi(x_i)\rangle = \langle U(x_i)|0\rangle \tag{3}$$

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Kernel Computation: Form the kernel K with entries:

$$K(i,j) = |\langle \varphi(x_i) | \varphi(x_j) \rangle|^2 \quad (4)$$

Encode New Sample: Compute the quantum state for new sample:

$$\langle |\varphi(x_{new})\rangle = \langle U(x_{new})|0\rangle \quad (5)$$

Classification: Evaluate the decision :

$$\int(x_{new}) = sign(\sum_{i=1}^{N} \propto_i y_i K(x_i, x_{new}) + b) \quad (6)$$

Intrusion Detected

Normal Activity =0

B. New Simulation Results

To further substantiate the algorithm, we ran additional simulations on a synthetic dataset of 1000 samples with 50 features per sample. The most important findings from these tests are as follows:

- F1-Score: The algorithm produced an F1-score of 97.5%, reflecting a very good precision-recall balance.
- ROC-AUC: An area which is included in the ROC curve (AUC) of 0.98 was witnessed, illustrating strong discriminative power.
- Processing Time: The mean classification time per sample was recorded at 0.015 seconds on an Intel i9 processor, which highlights the ability of the system to detect intrusion in real time.
- Robustness: The algorithm performed well under simulated adversarial attacks, confirming its resilience to changing attack modalities.

These new findings, along with the algorithm's effective incorporation of quantum computing methods, emphasize the promise of the QAI-ID system as a cutting-edge solution for securing smart vault environments in banking systems.

V. Workflow

1) Data Collection: Aggregate multiple data sources like security logs, IoT sensor readings and biometric authentication records from the smart vault system. Sensor fusion and completeness checking for data quality assurance.

2) Data Preprocessing: Once data has been collected, you will need to clean it by sorting out missing values, removing outliers, and normalizing features. Extract relevant attributes that characterize the temporal and spatial dimensions of vault action.

*3) Quantum Feature Encoding:*Convert the preprocessed vectors into quantum states using a parameterized unitary operator U(x). This is represented as:

$$\langle |\varphi(x)\rangle = \langle U(x)|0\rangle \quad (7)$$

allowing complex data patterns to be represented in high-dimensional Hilbert space.

4) Kernel Computation: Construct the quantum matrix K computing the main product between quantum states." Each element of matrix:

$$K(x_i, x_j) = |\langle \varphi(x_i) | \varphi(x_j) \rangle|^2 \quad (8)$$

5) QSVM Training: Use the kernel matrix to train a Quantum Support Vector Machine(QSVM). This is solved as an optimization problem to perceive the Lagrange multipliers and a bias term that separates normal activities from intrusions.

6) New Sample Encoding: For real-time analysis, encode new incoming data samples into quantum states using the same unitary transformation:

$$\langle |\varphi(x_{new})\rangle = \langle Ux_{new}|0\rangle \quad (9)$$

7) Classification: Classify new samples by evaluating the function that decides :

$$\int(x_{new}) = sign(\sum_{i=1}^{N} \propto_i y_i K(x_i, x_{new}) + b) \quad (10)$$

A negative value of $\int(x_{new})$ indicates an intrusion, while a positive value suggests normal activity.

8) Alert and Response: If an intrusion is detected, issue an alert and execute pre-defined response protocols. Keep log details of incident for rollback and tuning of the system.

9) Continuous Monitoring: Add the new data iteratively over time, being careful to use this data while continuing to monitor the systems. Continually retrain the model on new data (flow) to capture changing behavior patterns and overcome short-term memory.

VI. Literature Review

A. Intrusion Detection Systems (IDS) in Banking

Intrusion detection systems have a long history in cybersecurity. The early IDS research focused on both sig nature-based detection [1] and anomaly detection based on statistical approaches [2]. Over the growth of machine learning, various techniques have been adopted for identifying the intrusions in banking and financial systems, which include Support Vector Machines (SVM) [3], neural networks[4] and ensemble methods [5]. However, these conventional approaches are still limited with such advanced, high-dimensional security information in modern smart vault systems.

B. Quantum Computing and Machine Learning

By exploiting effects such as superposition and entanglement, quantum computing could offer a paradigm shift in how information is processed. For example, Shor's factorization algorithm for integers [6] and Grover's search algorithm [7] are early examples of quantum algorithms that are better than their classical counterparts. More recently, various quantum machine learning (QML) techniques, including quantum kernel estimation [8], and quantum neural networks[9] have been investigated for it's use in pattern recognition tasks. Quantum-aided algorithms have also been shown to discover non-linear trends in data at much higher Neyman-Pearson SDA than their classical counterparts [10] in all cases where the computationally expensive feature expansion is available option.

C. Quantum AI in Cybersecurity

The Application of quantum computing in cybersecurity is an upcoming research field. Quantum cryptographic

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protocols like Quantum Key Distribution (QKD) [11] have already been utilized to provide secure communication. In the case of IDS, researchers started exploring quantum algorithms for fast anomaly detection and threat mitigation [12]. The use Quantum feature maps with classical machine learning models, including SVMs, has been shown to enhance detection accuracy in a wide range of applications [13].

D. Vault Security Systems

Smart vaults are equipped with advanced technologies such as IoT sensors, biometric authentication, and automated access control to offer multi-layered protection. Current research has focused on developing ways to make such systems resilient against physical and cyber attacks [14]. The remaining problem is designing IDS able to process the heterogeneous data generated by smart vaults in real-time efficiently. Quantum AI-based solutions provide a likely remedy for avoiding such issues, with greater power to detect and withstand new avenues of attack.

E. Algorithm: QAID Training and Detection

Algorithm QAID Intrusion Detection

Input : Dataset $D = \{(x_i, y_i)\}$ for $i = 1,2, \dots, N$ and
Output: Trained QSVM for intrusion detection.

1) Preprocessing: For each sample x_i ,normalize features.
Perform feature selection/extraction.

2) Quantum Feature Encoding: For each sample x_i: Compute quantum state

$$\langle|\varphi(x_i)\rangle = \langle U(x_i)|0\rangle \tag{11}$$

3) Quantum Kernel Computation: For $i = 1\ to\ N$ For $j = 1\ to\ N$ Compute

$$K(x_i, x_j) = |\langle\varphi(x_i)|\varphi(x_j)\rangle|^2 \tag{12}$$

4) QSVM Training: Solve the quadratic optimization problem: Minimize

$$\frac{1}{2}\sum_{i,j} \propto_i \propto_j y_i y_j K(x_i x_j) - \sum_i \propto_i$$

Subject to:

$$\sum_i \propto_i y_i = 0$$

And

$$0 \leq \propto_i \leq C \tag{13}$$

Determine the bias term b.

5) Intrusion Detection: For a new data sample x_new:

- Compute
$$\langle|\varphi\rangle\langle x_new|\varphi\rangle = \langle U(x_new)|0\rangle \tag{14}$$
- For each support vector : Compute kernel value
$$K(x_i, x_{new}) = |\langle\varphi(x_i)|\varphi(x_{new})\rangle|^2 \tag{15}$$
- Compute
$$\int x_new = sign(\sum_i \propto_i y_i K(x_i, x_{new}) + b) \tag{16}$$
- If $\int x_new < 0$, flag as intrusion; otherwise, classify as normal.

6) Alert and Response: If an intrusion is detected, trigger the appropriate alert mechanisms.
Return: Trained QSVM model and real-time detection capability.

VII. Dataset and Experimental Setup

A. Real-Life Dataset

For experimental verification of our method, we employ a real-world dataset with security logs and sensor values from a bank's intelligent vault system. The dataset contains:

- Access Logs: Timestamped records of vault access events.
- Biometric Data: Authentication logs (fingerprint, facial recognition).
- Sensor Readings: Data from door sensors, motion detectors, and environmental sensors.
- Labeling: Each record is labeled as either "normal" or "intrusion" based on historical incident reports.

Example Features:
x_1: Time interval between successive access attempts.
x_2 : Authentication score (biometric match confidence).
x_3 : Sensor anomaly score (derived from sensor fusion algorithms).
. . .
x_d: Network traffic indicators related to vault operations.

B. Experimental Configuration

- Preprocessing: Missing values were imputed using mean imputation, and features were normalized to the range [0, 1].
- Parameter Setting:

a) Regularization parameter: C was optimized using cross-validation.

b) The unitary operator: $U(x)$ for quantum encoding was realized with a parameterized quantum circuit involving rotation gates.

- Evaluation Metrics: The system performance was measured based on accurate, precise answer, recalled, F1- score, and receiver operating characteristic (ROC) curve. These are defined below:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN'}$$

$$Precision = \frac{TP}{TP + FP'}$$

$$Recall = \frac{TP}{TP + FN'}$$

$$\text{F1-score} = \frac{2(Precision * Recall)}{Precision + Recall'} \tag{17}$$

- Implementation: Quantum elements were simulated with quantum computing libraries (Qiskit or Pennylane) whereas the classical elements were coded in Python with Scikit-Learn.

VIII. Results and Discussion

The experimental comparison proved that the QAID system provides a drastic improvement in accuracy of intrusion

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detection over the usual classical techniques. The findings include:

- Enhanced Detection Performance:

The QAID system yielded an accuracy rate of 96.5% with a accuracy of 95.2% and recall of 97.0% on the test set beating a SVM baseline by around 4–6% on each of the metrics.

- Robustness to High-Dimensional Data:

The quantum kernel approach effectively encoded intricate nonlinear associations in the security data, which was confirmed through enhanced performance on high-dimensional subsets of features.

- Real-Time Detection Capability:

Even though the quantum feature encoding and kernel computation are computationally expensive, hybrid implementations and circuit optimizations made near- real-time detection appropriate for operational settings.

- Scalability Considerations:

Simulation outcomes indicate that as the quantum hard- ware is developed, the QAID system can be increased to larger-sized datasets and even more sophisticated smart vault environments.

IX. Conclusion and Future Work

A new Quantum AI-based Intrusion Detection System specifically designed for smart vault security in financial systems. By using quantum feature encoding and quantum kernel techniques within a QSVM paradigm, our method shows significant improvements compared to traditional approaches in identifying intricate intrusion patterns. Future work will involve:

- Deploying the system on near-term quantum hardware to evaluate performance in real-world conditions.
- Scaling the model to support more modalities of additional data (live video analytics).
- Investigation of hybrid quantum-classical optimisation methods further lowering computational latency.
- Testing of the system with adversarial attacks to improve the system's resistance.

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